

A Gauss-Stokes boundary integral framework for consistent surface orientation

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Abstract

Consistent orientation of a closed surface is globally constrained and cannot in general be determined from local information alone. The classical Gauss identity provides a linear global condition, but it is non-injective when the unknown is a surface vector field. We augment this identity with exact homogeneous constraints generated by the fundamental solution of the incompressible Stokes equations. The resulting Stokes single-layer boundary operator has a classical and explicit nullspace. For a boundary with M connected components, its kernel is precisely the M -dimensional span of the component normals. We prove that the Gauss operator restricted to this nullspace is an invertible triangular containment matrix determined by surface nesting. Consequently, the coupled Gauss-Stokes system uniquely recovers the outward normal for disconnected domains and arbitrarily nested cavities. We further establish a Lipschitz stability estimate in the natural Sobolev scale.

Keywords: surface orientation, boundary integral equations, Stokes single-layer potential, Gauss formula, stability

1. Introduction

Consistent surface orientation is a basic requirement in geometric modeling, reconstruction, volumetric reasoning, and boundary-based numerical methods. Local normal directions can usually be estimated from neighborhoods, but

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their signs are not determined locally. The difficulty becomes more pronounced for disconnected components, nested shells, cavities, thin structures, and noisy samples. Classical propagation methods orient local normals along neighborhood graphs (Hoppe et al., 1992; König and Gumhold, 2009; Liu et al., 2014), while graph-energy and parallel formulations seek more global consistency (Scherer et al., 2017; Jakob et al., 2019). Recent methods use winding numbers, Poisson equations, Gauss identities, Green identities, stochastic fields, volumetric partitioning, or global optimization to inject nonlocal information (Metzer et al., 2021; Xu et al., 2023; Lin et al., 2022; Xiao et al., 2023; Liu et al., 2024; Lin et al., 2024; Huang et al., 2024; Liu et al., 2025; Fu et al., 2025; Li et al., 2025; Ge et al., 2026; Wang et al., 2026; Wu et al., 2026).

This paper studies a boundary integral system for determining a consistent normal orientation. The classical Gauss identity provides a natural starting point. Let

$$\Phi(z) = \frac{1}{4\pi|z|}.$$

For the outward normal n_Ω of a smooth bounded domain, the averaged boundary trace of the Laplace double-layer representation gives

$$-\text{p.v.} \int_{\partial\Omega} n_\Omega(y) \cdot \nabla_y \Phi(x - y) \, dS_y = \frac{1}{2}, \quad x \in \partial\Omega.$$

Here p.v. denotes the principal value. The corresponding trace and jump relations for Laplace layer potentials are classical (Costabel, 1988; McLean, 2000; Hsiao and Wendland, 2008). The identity gives linear constraints on the outward normal and depends only on positions and unsigned area measure. It is therefore attractive for unoriented geometry, and it underlies Parametric Gauss Reconstruction (PGR) (Lin et al., 2022). However, as an equation for a general three-component boundary field, it is not injective. In fact, divergence-free tangential fields form an infinite-dimensional nullspace. A Gauss formulation must consequently add a selection principle, such as minimum-norm regularization, or supplement the system with additional equations.

Here we take the second route and add exact homogeneous constraints. The divergence theorem shows that every divergence-free vector field q produces

$$\int_{\partial\Omega} q \cdot n_\Omega \, dS = \int_\Omega \nabla \cdot q(x) \, dx = 0.$$

The incompressible Stokes fundamental solution provides a canonical family of such divergence-free vector fields. This leads to the Stokes single-layer constraint

$$\mathcal{V}n_\Omega := \int_\Gamma E(x-y)n_\Omega(y) \, dS_y = 0.$$

where $E(z)$ is the Stokeslet (Pozrikidis, 1992)

$$E(z) = \frac{1}{8\pi} \left(\frac{I}{|z|} + \frac{z \otimes z}{|z|^3} \right), \quad z \neq 0.$$

The key advantage is not merely the availability of more equations. Classical Stokes boundary-integral theory gives an explicit geometric nullspace: for M closed boundary components,

$$\ker \mathcal{V} = \text{span}\{\eta_1, \dots, \eta_M\},$$

where η_j is the geometric normal of the j -th connected component. The connected-boundary normal kernel and the single-layer jump structure are stated explicitly in Kohr et al. (2026, Thms. 5.12 and 6.10); the multiply connected case and the associated factor-space/ellipticity issue are treated by Reidinger and Steinbach (2003) and Steinbach (2008, Sec. 6.8). Thus the Stokes constraint reduces an infinite-dimensional vector-field ambiguity to one scalar normal mode per component. The remaining coefficients are then determined by the Gauss identity, whose restriction to the Stokes kernel is an invertible triangular matrix encoding surface nesting.

The boundary integral system studied in this paper couples the Gauss identity and the Stokes single-layer constraint. We prove that the Gauss–Stokes system has a unique solution equal to the outward normal, including for disconnected domains and arbitrarily nested cavities. Moreover, the same decomposition yields a natural stability estimate: the Stokes constraint controls the complement of its finite-dimensional kernel, while the Gauss identity controls the kernel itself.

The main contributions of this work are threefold:

1. We introduce the Gauss–Stokes system, a boundary integral formulation that couples the classical Gauss identity with homogeneous constraints induced by the Stokes single-layer operator. The Stokes constraint exploits the geometric structure of its nullspace, reducing the infinite-dimensional ambiguity of the unknown surface vector field to a finite-dimensional space spanned by the normal modes of the connected boundary components.

2. We establish the existence and uniqueness of the solution to the Gauss-Stokes system. In particular, we show that the outward normal satisfies the coupled system and prove that it is the unique solution. The key ingredient is that the Gauss operator, when restricted to the nullspace of the Stokes single-layer operator, is represented by an invertible containment matrix determined by the nesting relations among boundary components. The result applies to both disconnected domains and arbitrarily nested cavities.
3. We develop a stability analysis of the Gauss-Stokes system in the natural Sobolev setting. By combining the coercive control of the Stokes operator on the complement of its kernel with the finite-dimensional control provided by the Gauss operator on the kernel, we derive a bounded-below estimate for the coupled operator and a Lipschitz stability bound.

2. Related work

2.1. Global normal orientation and unoriented reconstruction

The seminal work of Hoppe et al. (1992) propagates local normal signs through a minimum-spanning tree. König and Gumhold (2009) analyze and improve consistent graph propagation, while the multi-source strategy of Liu et al. (2014) is designed to repair propagation failures. Global graph-energy optimization was developed for large point clouds by Schertler et al. (2017) and subsequently parallelized for raw unorganized data by Jakob et al. (2019). Thin structures motivated the offset-surface/global-transport strategy of Xu et al. (2018). Harmonic point-cloud orientation provided another early global alternative based on smooth functions of the point set (Seversky et al., 2011), and fast winding-number evaluation supplied an efficient volumetric primitive for soups and clouds (Barill et al., 2018).

More recent methods improve robustness by introducing learned or analytic global fields and objectives: dipole propagation combines learned local patches with a global dipole field (Metzger et al., 2021); GCNO and WNNC regularize winding-number consistency (Xu et al., 2023; Lin et al., 2024); boundary integration and diffusing winding gradients provide further global mechanisms (Liu et al., 2024, 2025). Stochastic Normal Orientation (SNO) introduces a probabilistic global/local objective (Huang et al., 2024), while FaCE uses an electrostatic Faraday-cage model (Scrivener et al., 2025). Delaunay-graph least squares and divide-and-conquer integer optimization address difficult topology and non-watertight scene-scale inputs (Fu et al., 2025; Li et al., 2025).

Boundary-aware narrow-band partitioning targets noisy and non-watertight inputs (Ge et al., 2026), and robust spherical fitting coupled with a variational propagation model jointly estimates and globally orients normals (Wang et al., 2026). Very recent work also emphasizes scale and efficiency, including GPU-accelerated SNO (Zhang et al., 2026) and MegaNorm for super-large point sets (Li et al., 2026).

2.2. Gauss identity and PDE-based formulations

Modified Gauss formulas have been used to reconstruct implicit surfaces from unoriented data (Lu et al., 2019). PGR parameterizes the Gauss formula by unknown oriented surface elements and resolves the resulting underdetermined linear system through a minimum-norm or regularized selection (Lin et al., 2022). Poisson-based approaches jointly optimize implicit functions and normals by adding isovalue constraints (Xiao et al., 2023). More recently, CAGR augments the Gauss formula with finite convection vectors (Ma et al., 2025b), and AGR uses anisotropic Laplace operators to create additional constraints and a stable linear system (Ma et al., 2025a). These works establish equation augmentation as a powerful design principle.

2.3. Homogeneous constraints and boundary integral operators

Recent work by Ma et al. (2026) constructs divergence-free function fields and applies the divergence theorem to add homogeneous constraints. G3R uses Green’s third identity to remove the tangential nullspace of winding-number formulations (Wu et al., 2026). Gotsman and Hormann derive a linear orientation method from the classical Stokes theorem (Gotsman and Hormann, 2024). This is mathematically distinct from our use of the incompressible Stokes PDE and its fundamental solution.

For the analytic facts used below, we distinguish the Laplace and Stokes layer-potential theories. For Laplace operators, continuity, Sobolev mapping, jump relations, and ellipticity of classical layer potentials are established in standard boundary-integral references (Costabel, 1988; McLean, 2000; Hsiao and Wendland, 2008; Sauter and Schwab, 2011). For the Stokes system, the velocity single layer, its trace/traction jumps, variational formulation, and factor-space coercivity are treated in Pozrikidis (1992), Kohr and Wendland (2006), and Steinbach (2008, Sec. 6.8). In particular, Reidinger and Steinbach (2003) treats multiply connected domains, where topology forces quotient/factor spaces for the single-layer operator. A recent layer-potential

analysis makes the no-jump velocity trace, elliptic order (-1) symbol, self-adjoint Fredholm property, traction jump, and connected-boundary normal kernel explicit (Kohr et al., 2026, Thms. 5.12, 5.13, and 6.10). These are precisely the operator properties used in our nullspace and stability arguments.

3. Geometric setting and the ambiguity of the Gauss identity

Let $\Omega \subset \mathbb{R}^3$ be a bounded open set whose boundary is the disjoint union

$$\Gamma = \partial\Omega = \bigsqcup_{j=1}^M \Sigma_j, \quad (1)$$

where every Σ_j is a connected, embedded, smooth, closed surface.

By the Jordan-Brouwer separation theorem, each Σ_j encloses a bounded region D_j . Let m_j denote the unit normal on Σ_j pointing outward from D_j . This is the geometric outward normal of the individual closed surface. It need not agree with the outward normal of Ω . For example, if Σ_j is a cavity boundary, then

$$n_\Omega|_{\Sigma_j} = -m_j.$$

Extend m_j to the full boundary by

$$\eta_j(x) = \begin{cases} m_j(x), & x \in \Sigma_j, \\ 0, & x \in \Gamma \setminus \Sigma_j. \end{cases} \quad (2)$$

The operators introduced below depend only on $x, y \in \Gamma$ and on the unsigned surface measure dS . Hence the assembly of the continuous system does not require m_j or n_Ω . The normals enter only in the analysis.

For two distinct components define

$$\Sigma_k \prec \Sigma_j \iff \Sigma_k \subset D_j.$$

Since the surfaces are pairwise disjoint, this relation defines a forest-like nesting structure. The depth of Σ_j is

$$d_j = \#\{i \neq j : \Sigma_j \subset D_i\}. \quad (3)$$

Because Ω is bounded, membership in Ω alternates whenever one crosses a boundary component while moving inward from the unbounded exterior. Consequently,

$$n_\Omega|_{\Sigma_j} = (-1)^{d_j} m_j. \quad (4)$$

Thus outermost components have positive sign, first-level cavities have negative sign, islands inside cavities have positive sign again, and so forth.

For $s \in \mathbb{R}$ we use

$$H^s(\Gamma)^3 = \bigoplus_{j=1}^M H^s(\Sigma_j)^3.$$

The natural density space for the Stokes single-layer operator is

$$X := H^{-1/2}(\Gamma)^3.$$

3.1. Gauss identity

Let

$$\Phi(z) = \frac{1}{4\pi|z|}.$$

For the outward unit normal n_Ω of a bounded smooth domain, the Gauss formula gives

$$-\int_{\partial\Omega} n_\Omega(y) \cdot \nabla_y \Phi(x-y) \, dS_y = \begin{cases} 1, & x \in \Omega, \\ 0, & x \notin \overline{\Omega}. \end{cases}$$

Taking the averaged boundary limit using the classical Laplace double-layer jump relation (Costabel, 1988; McLean, 2000; Hsiao and Wendland, 2008) motivates the orientation operator

$$(\mathcal{B}\varphi)(x) = -\text{p.v.} \int_{\Gamma} \varphi(y) \cdot \nabla_y \Phi(x-y) \, dS_y, \quad x \in \Gamma, \quad (5)$$

for which

$$\mathcal{B}n_\Omega = \frac{1}{2}\mathbf{1}_\Gamma. \quad (6)$$

Equation (6) is attractive because it is linear in the unknown vector field and its kernel can be assembled from positions and unsigned surface measure alone. Nevertheless, it is not an injective equation for a general three-component surface vector field.

Let τ be a smooth tangential vector field on Γ . Since

$$\tau \cdot \nabla_y \Phi(x-y) = \tau \cdot \nabla_\Gamma \Phi(x-y),$$

integration by parts gives, in the sense of principal-value,

$$\mathcal{B}\tau = \mathcal{S}_\Delta(\operatorname{div}_\Gamma \tau), \quad (7)$$

where

$$(\mathcal{S}_\Delta f)(x) = \int_\Gamma \Phi(x - y) f(y) \, dS_y$$

is the scalar Laplace single-layer operator. The identity is compatible with the standard Sobolev mapping of the Laplace single layer (Costabel, 1988; McLean, 2000; Hsiao and Wendland, 2008).

Hence, for every tangential divergence-free field τ satisfying

$$\tau \cdot n_\Omega = 0, \quad \operatorname{div}_\Gamma \tau = 0,$$

we obtain

$$\mathcal{B}\tau = 0.$$

Consequently, the affine solution set of Eq. (6) contains

$$n_\Omega + \mathcal{T}_0, \quad \mathcal{T}_0 = \{\tau \in C^\infty(T(\Gamma)) : \operatorname{div}_\Gamma \tau = 0\},$$

and is therefore nonunique in the linear vector-field formulation.

To remove this ambiguity, Parametric Gauss Reconstruction (PGR) (Lin et al., 2022) parametrizes the Gauss formula by unknown linear surface elements and solves the resulting underdetermined dense system using a regularized strategy. This is a natural and useful computational choice. A strictly convex regularizer could generate a unique solution, but the resulting uniqueness then depends on the regularizer. This motivates adding more exact equations before invoking regularization. CAG, AGR, the wavelet method, and G3R pursue related goals through different augmented PDE or Green-identity constructions (Ma et al., 2025b,a, 2026; Wu et al., 2026).

4. Gauss-Stokes boundary integral formulation

4.1. Divergence-free constraints

Let $q : \Omega \rightarrow \mathbb{R}^3$ be sufficiently smooth and divergence-free:

$$\operatorname{div} q = 0.$$

The divergence theorem gives

$$\int_{\Gamma} q(y) \cdot n_{\Omega}(y) \, dS_y = 0. \quad (8)$$

Thus every divergence-free vector field provides a homogeneous linear equation for the unknown normal. This general augmentation principle is also used in recent wavelet-based unoriented reconstruction (Ma et al., 2026). In this paper, we focus on the particular divergence-free family supplied by the incompressible Stokes system.

4.2. Stokeslet augmentation

Consider the steady incompressible Stokes equations

$$-\Delta u + \nabla p = f, \quad \nabla \cdot u = 0. \quad (9)$$

The velocity fundamental solution is the Stokeslet (Pozrikidis, 1992, Ch. 2)

$$E(z) = \frac{1}{8\pi} \left(\frac{I}{|z|} + \frac{z \otimes z}{|z|^3} \right), \quad z \neq 0. \quad (10)$$

Its incompressibility property implies

$$\operatorname{div}_y E(x - y) = 0. \quad (11)$$

Hence,

$$\int_{\Gamma} E(x - y) n_{\Omega}(y) \, dS_y = 0. \quad (12)$$

Taking the continuous boundary trace defines the Stokes single-layer boundary operator

$$(\mathcal{V}\varphi)(x) = \int_{\Gamma} E(x - y) \varphi(y) \, dS_y, \quad x \in \Gamma, \quad (13)$$

and therefore

$$\mathcal{V}n_{\Omega} = 0. \quad (14)$$

4.3. Coupled formulation

Combining the nonhomogeneous Gauss identity with the homogeneous incompressibility constraints gives the coupled system in Eq. (15).

$$\mathcal{V}\varphi = 0, \quad \mathcal{B}\varphi = \frac{1}{2}. \quad (15)$$

The Stokes equation removes the non-normal freedom of the Laplace equation but retains a finite-dimensional normal kernel. The Gauss identity then fixes the coefficients inside the null space.

Conceptually, this replaces part of the regularization burden by exact Stokes constraints. Regularization remains useful under sampling error or noise, but it is no longer the mechanism that creates continuum identifiability.

4.4. Stokes single-layer operator

The pressure kernel associated with the Stokeslet is

$$\Pi(z) = \frac{z}{4\pi|z|^3}. \quad (16)$$

For a vector density φ define the Stokes single-layer velocity and pressure potentials

$$(\mathcal{S}\varphi)(x) = \int_{\Gamma} E(x-y)\varphi(y) \, dS_y, \quad (17)$$

$$(\mathcal{P}\varphi)(x) = \int_{\Gamma} \Pi(x-y) \cdot \varphi(y) \, dS_y. \quad (18)$$

In $\mathbb{R}^3 \setminus \Gamma$ these satisfy the homogeneous Stokes system.

For a smooth closed boundary, the Stokes velocity single-layer potential has no jump across Γ , so its common trace is exactly the operator $\mathcal{V}$ defined in Eq. (13). The corresponding one-sided tractions have the standard density jump. These trace and traction relations are classical, and are stated explicitly in Kohr et al. (2026, Thm. 5.12), also in Pozrikidis (1992, Ch. 2) and Kohr and Wendland (2006). Its kernel is weakly singular, of order $O(|x-y|^{-1})$, and therefore integrable on a two-dimensional smooth surface.

The Sobolev mapping property is

$$\mathcal{V} : H^{-1/2}(\Gamma)^3 \longrightarrow H^{1/2}(\Gamma)^3 \quad (19)$$

boundedly (Steinbach, 2008, Sec. 6.8). In the smooth-boundary pseudodifferential calculus, $\mathcal{V}$ is elliptic of order -1 with self-adjoint principal symbol (Kohr et al., 2026, Thm. 5.12). It is Fredholm of index zero and self-adjoint on L^2 space (Kohr et al., 2026, Thm. 5.13). Its energy form is nonnegative, and after removing the normal kernel it is coercive in the natural $H^{-1/2}$ scale (Steinbach, 2008, Thm. 6.39). The multiply connected stabilization/factor-space setting is developed in Reidinger and Steinbach (2003).

For every component normal η_j defined in Eq. (2), the same divergence-free argument as in Eq. (12) yields

$$\mathcal{V}\eta_j = 0. \quad (20)$$

Thus disconnectedness immediately produces at least one null vector per connected boundary component. The next section shows that there are no others.

4.5. Nullspace characterization

For a connected smooth boundary, the normal kernel is the classical Stokes single-layer kernel (Kohr et al., 2026, Thm. 6.10). For multiply connected domains, one normal mode occurs for each closed boundary component; see Reidinger and Steinbach (2003) and Steinbach (2008, Sec. 6.8). We record the precise form needed for the nested-boundary geometry and give a short proof.

Theorem 4.1 (Nullspace structure). *Let Γ be as in Eq. (1). Then*

$$\ker \mathcal{V} = K := \text{span}\{\eta_1, \dots, \eta_M\}, \quad \dim K = M. \quad (21)$$

Proof. By Eq. (20),

$$K \subseteq \ker \mathcal{V}.$$

Conversely, let $\varphi \in \ker \mathcal{V}$ and set

$$u = \mathcal{S}\varphi, \quad p = \mathcal{P}\varphi.$$

Since the Stokes single-layer velocity is continuous across the boundary,

$$u|_{\Gamma} = \mathcal{V}\varphi = 0.$$

Consider any connected component U_ℓ of $\mathbb{R}^3 \setminus \Gamma$. In U_ℓ ,

$$-\Delta u + \nabla p = 0, \quad \nabla \cdot u = 0,$$

with homogeneous Dirichlet trace. In the unbounded component the single-layer field also satisfies the standard decay condition. Uniqueness for the homogeneous Stokes Dirichlet problem (Galdi, 2011; Kohr et al., 2026) implies

$$u \equiv 0 \quad \text{in every } U_\ell.$$

Hence $\nabla p = 0$, so p is constant on each U_ℓ .

Across a component Σ_j , the Stokes single-layer traction satisfies the standard jump relation (Kohr et al., 2026, Thm. 5.12). Since the velocity vanishes on both sides, the viscous part of the stress vanishes. The traction jump is therefore generated only by the difference of the two constant pressures. With either of the equivalent standard traction sign conventions, it follows that

$$\varphi|_{\Sigma_j} = \alpha_j m_j$$

for some scalar α_j . Thus

$$\varphi = \sum_{j=1}^M \alpha_j \eta_j \in K.$$

Because the supports of the η_j are disjoint, they are linearly independent. Hence $\dim K = M$. $\square$

5. Uniqueness of Gauss-Stokes system

Recall the Gauss orientation operator $\mathcal{B}$ from Eq. (5). For smooth Γ , its normal part is the classical Laplace double-layer boundary operator and its tangential part is represented by Eq. (7). Standard layer-potential mapping theory therefore gives

$$\mathcal{B} : H^{-1/2}(\Gamma)^3 \longrightarrow H^{-1/2}(\Gamma) \quad (22)$$

boundedly (Costabel, 1988; McLean, 2000; Hsiao and Wendland, 2008).

Proposition 5.1 (Containment matrix). *For $x \in \Sigma_k$,*

$$(\mathcal{B}\eta_j)(x) = \begin{cases} \frac{1}{2}, & j = k, \\ 1, & j \neq k \text{ and } \Sigma_k \subset D_j, \\ 0, & \Sigma_k \not\subset \overline{D_j}. \end{cases} \quad (23)$$

Proof. If $x \notin \Sigma_j$, the divergence theorem gives

$$\begin{aligned} - \int_{\Sigma_j} m_j(y) \cdot \nabla_y \Phi(x - y) \, dS_y &= - \int_{D_j} \Delta_y \Phi(x - y) \, dy \\ &= \begin{cases} 1, & x \in D_j, \\ 0, & x \notin \overline{D_j}. \end{cases} \end{aligned}$$

As x approaches Σ_j , the classical Laplace double-layer jump relation gives the averaged boundary value $1/2$ with the present sign convention (Costabel, 1988; McLean, 2000; Hsiao and Wendland, 2008). $\square$

Let

$$k = \sum_{j=1}^M \alpha_j \eta_j \in K, \quad \alpha = (\alpha_1, \dots, \alpha_M)^T.$$

The function $\mathcal{B}k$ is constant on each Σ_k . Hence $\mathcal{B}|_K$ is represented by the matrix

$$T_{kj} = \begin{cases} \frac{1}{2}, & k = j, \\ 1, & k \neq j, \Sigma_k \subset D_j, \\ 0, & \text{otherwise.} \end{cases} \quad (24)$$

Order the components by nondecreasing nesting depth. Then T is triangular and

$$\det T = 2^{-M} \neq 0. \quad (25)$$

Corollary 5.2 (Invertibility on the Stokes null space). *The restriction*

$$\mathcal{B}|_K : K \longrightarrow \mathcal{B}(K)$$

is injective. Equivalently, there exists $\beta_\Gamma > 0$ such that

$$\|k\|_{H^{-1/2}(\Gamma)^3} \leq \beta_\Gamma^{-1} \|\mathcal{B}k\|_{H^{-1/2}(\Gamma)}, \quad k \in K. \quad (26)$$

Proof. The matrix T is invertible by Eq. (25). Since K is finite dimensional, all norms on K are equivalent, which yields Eq. (26). $\square$

From Eq. (4),

$$n_\Omega = \sum_{j=1}^M (-1)^{d_j} \eta_j. \quad (27)$$

The coefficient vector satisfies

$$T \begin{pmatrix} (-1)^{d_1} \\ \vdots \\ (-1)^{d_M} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}. \quad (28)$$

Thus the Gauss identity not only fixes the scale on each component, but also detects the sign reversal on cavity boundaries.

We now prove that the Gauss-Stokes system Eq. (15) is exactly identifiable.

Theorem 5.3 (Uniqueness of the Laplace–Stokes orientation system). *Let $\Omega \subset \mathbb{R}^3$ be bounded and let $\Gamma = \partial\Omega$ be the finite disjoint union of connected smooth closed surfaces. Then Eq. (15) has a unique solution in $H^{-1/2}(\Gamma)^3$, namely*

$$\varphi = n_\Omega. \quad (29)$$

The result allows disconnected domains, cavities, and arbitrary finite nesting depth.

Proof. First, n_Ω is a solution. Since the Stokeslet is divergence-free in the source variable,

$$\mathcal{V}n_\Omega(x) = \int_{\partial\Omega} E(x-y)n_\Omega(y) \, dS_y = \int_{\Omega} \operatorname{div}_y E(x-y) \, dy = 0.$$

Likewise, the Laplace Green identity gives

$$-\int_{\partial\Omega} n_\Omega(y) \cdot \nabla_y \Phi(x-y) \, dS_y = \begin{cases} 1, & x \in \Omega, \\ 0, & x \notin \overline{\Omega}, \end{cases}$$

and the averaged boundary limit yields

$$\mathcal{B}n_\Omega = \frac{1}{2}.$$

Now let φ satisfy Eq. (15). By Theorem 4.1,

$$\varphi \in K, \quad \varphi = \sum_{j=1}^M \alpha_j \eta_j.$$

The Laplace equation is equivalent to

$$T\alpha = \frac{1}{2}\mathbf{1}.$$

Since T is invertible, the coefficient vector is unique. Because n_Ω is already a solution, the unique coefficient vector is the one in Eq. (27). Hence $\varphi = n_\Omega$. $\square$

6. Stability analysis

One important fact to analyze the stability is that modulo the finite-dimensional space K , the operator $\mathcal{V}$ is elliptic/coercive of order -1 . In the connected case this is the content of the Stokes single-layer ellipticity estimate (Steinbach, 2008, Thm. 6.39). For multiple boundary components the corresponding factor-space stabilization is treated in Reidinger and Steinbach (2003). Equivalently, there exists $C_V > 0$ such that

$$\text{dist}_{H^{-1/2}}(\varphi, K) \leq C_V \|\mathcal{V}\varphi\|_{H^{1/2}(\Gamma)^3}, \quad \varphi \in H^{-1/2}(\Gamma)^3. \quad (30)$$

This is the infinite-dimensional part of the stability mechanism.

Theorem 6.1 (Stability estimate). *There exists a constant $C_\Gamma > 0$, depending only on the fixed geometry (under the unit-viscosity normalization), such that every $\varphi \in H^{-1/2}(\Gamma)^3$ satisfies*

$$\|\varphi\|_{H^{-1/2}(\Gamma)^3} \leq C_\Gamma [\|\mathcal{V}\varphi\|_{H^{1/2}(\Gamma)^3} + \|\mathcal{B}\varphi\|_{H^{-1/2}(\Gamma)}]. \quad (31)$$

Proof. By Eq. (30), choose $k \in K$ such that

$$r := \varphi - k$$

satisfies

$$\|r\|_{H^{-1/2}} \leq C \|\mathcal{V}\varphi\|_{H^{1/2}}, \quad (32)$$

because $\mathcal{V}k = 0$.

Next,

$$\mathcal{B}k = \mathcal{B}\varphi - \mathcal{B}r.$$

From Corollary 5.2,

$$\|k\|_{H^{-1/2}} \leq C_K \|\mathcal{B}k\|_{H^{-1/2}}, \quad k \in K. \quad (33)$$

Using boundedness of

$$\mathcal{B} : H^{-1/2}(\Gamma)^3 \rightarrow H^{-1/2}(\Gamma)$$

together with Eq. (33),

$$\|k\|_{H^{-1/2}} \leq C_K \|\mathcal{B}k\|_{H^{-1/2}} \leq C_K (\|\mathcal{B}\varphi\|_{H^{-1/2}} + \|\mathcal{B}\| \|r\|_{H^{-1/2}}).$$

Combining this with Eq. (32) and

$$\|\varphi\| \leq \|r\| + \|k\|$$

gives Eq. (31). □

Corollary 6.2. *For any approximation $\tilde{n} \in H^{-1/2}(\Gamma)^3$,*

$$\|\tilde{n} - n_\Omega\|_{H^{-1/2}} \leq C_\Gamma \left[\|\mathcal{V}\tilde{n}\|_{H^{1/2}} + \left\| \mathcal{B}\tilde{n} - \frac{1}{2} \right\|_{H^{-1/2}} \right]. \quad (34)$$

Proof. Apply Theorem 6.1 to

$$e = \tilde{n} - n_\Omega$$

and use

$$\mathcal{V}n_\Omega = 0, \quad \mathcal{B}n_\Omega = \frac{1}{2}.$$

□

For computation, an analogous estimate in unscaled L^2 norms would be attractive, but it is not available uniformly across spatial frequencies. In general there is no constant C only depends on the surface for an estimate of the form

$$\|\varphi\|_{L^2(\Gamma)^3} \leq C \left(\|\mathcal{V}\varphi\|_{L^2(\Gamma)^3} + \|\mathcal{B}\varphi\|_{L^2(\Gamma)} \right). \quad (35)$$

Indeed, on the infinite-dimensional tangential divergence-free subspace contained in $\ker \mathcal{B}$, the Stokes single layer remains and order -1 pseudodifferential operator and therefore smooths high spatial frequencies (Kohr et al., 2026, Thms. 5.12–5.13). The Laplace equation removes the finite-dimensional kernel of $\mathcal{V}$, but not this high-frequency smoothing. Consequently, a discretization measured in unscaled Euclidean residual norms should be expected to become increasingly ill-conditioned under refinement. Numerical implementations require the appropriate preconditioning or regularization.

7. Relation to convection-augmented Gauss reconstruction

Convection Augmented Gauss Reconstruction (CAGR) (Ma et al., 2025b) enriches the classical isotropic Gauss formula by introducing a constant convection vector. The convection family also provides an illuminating alternative derivation of the Gauss–Stokes system. This relation helps explain why CAGR’s finite set of convection directions supplies information closely related to the homogeneous Stokes constraints derived below.

7.1. A convection-augmented Green identity

For a constant vector $c \in \mathbb{R}^3$, consider

$$L_c = \Delta - c \cdot \nabla. \quad (36)$$

Since c is constant,

$$L_c u = \operatorname{div}(\nabla u - cu).$$

Let $z = y - x$, $r = |z|$, and define

$$G_c(y, x) = -\frac{1}{4\pi r} \exp\left(\frac{1}{2}c \cdot z - \frac{1}{2}|c|r\right). \quad (37)$$

Then

$$L_c^y G_c(y, x) = \delta_x(y).$$

Applying the divergence theorem to the flux

$$\nabla_y G_c - c G_c$$

and taking the averaged boundary limit yields

$$\frac{1}{2} = \int_{\Gamma} n_{\Omega}(y) \cdot (\nabla_y G_c(y, x) - c G_c(y, x)) \, dS_y, \quad x \in \Gamma. \quad (38)$$

7.2. Zero convection recovers the Gauss identity

At $c = 0$,

$$G_0(y, x) = -\frac{1}{4\pi|x-y|} = -\Phi(x-y).$$

Therefore Eq. (38) becomes

$$-\text{p. v.} \int_{\Gamma} n_{\Omega}(y) \cdot \nabla_y \Phi(x-y) \, dS_y = \frac{1}{2},$$

namely

$$\mathcal{B}n_{\Omega} = \frac{1}{2}.$$

Thus the classical Laplace Gauss equation is the zero-convection member of the augmented family.

7.3. *The odd infinitesimal variation generates the Stokes constraints*

Fix $a \in \mathbb{R}^3$ and set

$$c = \pm ta, \quad t > 0.$$

Let

$$F_c(y, x) = \nabla_y G_c(y, x) - c G_c(y, x).$$

The right-hand side of Eq. (38) is the same for ta and $-ta$, hence

$$0 = \int_{\Gamma} n_{\Omega}(y) \cdot \frac{F_{ta}(y, x) - F_{-ta}(y, x)}{t} \, dS_y. \quad (39)$$

The screening factor

$$\exp\left(-\frac{1}{2}|c|r\right)$$

is even under $c \mapsto -c$ and does not contribute to the odd first variation. With

$$G_0(z) = -\frac{1}{4\pi r},$$

we have

$$\frac{G_{ta} - G_{-ta}}{t} \longrightarrow (a \cdot z)G_0,$$

and

$$\frac{-ta G_{ta} - (+ta)G_{-ta}}{t} \longrightarrow -2aG_0.$$

Therefore

$$\begin{aligned} \lim_{t \downarrow 0} \frac{F_{ta} - F_{-ta}}{t} &= \nabla_z((a \cdot z)G_0) - 2aG_0 \\ &= -aG_0 + (a \cdot z)\nabla_z G_0 \\ &= \frac{1}{4\pi} \left(\frac{I}{r} + \frac{z \otimes z}{r^3} \right) a. \end{aligned} \quad (40)$$

Substituting into Eq. (39) and using arbitrary a gives

$$\int_{\Gamma} \left(\frac{I}{|x-y|} + \frac{(x-y) \otimes (x-y)}{|x-y|^3} \right) n_{\Omega}(y) \, dS_y = 0. \quad (41)$$

Multiplying by the positive unit-viscosity factor $1/(8\pi)$ gives precisely

$$\mathcal{V}n_{\Omega} = 0.$$

The preceding calculation gives a useful interpretation of the proposed coupling. CAGR samples the nonlinear family of convection-augmented Gauss equations at finitely many nonzero velocities. In contrast, the map

$$a \mapsto \left. \frac{d}{dt} \right|_{t=0^+} (F_{ta} - F_{-ta})$$

is linear in the signed direction a , and its matrix representation is, up to a scalar factor, precisely the Stokeslet. Hence the Stokes constraint collects the complete first-order odd directional information at $c = 0$ without selecting a finite set of convection directions.

8. Conclusion

In this paper, we introduced the Gauss-Stokes system, a coupled boundary integral formulation for the recovery and characterization of consistently oriented outward normal fields on closed surfaces. The central idea is to combine the Gauss identity with homogeneous constraints generated by the Stokes single-layer operator. While the Gauss relation alone does not eliminate all ambiguities, the Stokes operator restricts the remaining freedom to a finite-dimensional nullspace associated with the connected components of the boundary.

We established the existence and uniqueness of solutions to the Gauss-Stokes system and proved that its unique solution is the outward unit normal. The argument also reveals how the topology and nesting structure of the geometry enter the problem. In particular, the action of the Gauss operator on the nullspace of the Stokes operator reduces to a finite-dimensional system determined by the containment relations among boundary components. This allows the uniqueness result to cover both disconnected geometries and configurations with arbitrarily nested cavities within a unified framework.

We also analyzed the stability of the Gauss-Stokes system in the natural Sobolev spaces. The proof separates the unknown into its nullspace component and its complementary component. The Stokes operator provides coercive control on the latter, whereas the Gauss operator controls the former through the finite-dimensional containment structure. Combining these two mechanisms yields a stability estimate for the coupled operator and, consequently, a Lipschitz stability estimate that bounds the error in the recovered normal field by the residual of the Gauss-Stokes system.

The proposed Gauss-Stokes formulation provides a mathematically well-posed framework for determining consistently oriented surface normals on geometrically complex boundaries. The results provide a theoretical basis for the development and analysis of numerical methods based on the proposed system.

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